

Theoretical Appendix: Protocol for Active Vacuum Expansion

Author: Peter Kyjovsky

1. Problem

Natural gaps (interstitials) in the Higgs Lattice (Mackay cluster structure) are of subatomic dimensions. For macroscopic utilization (inertialess transport), it is necessary to enlarge these gaps.

2. The Vacuum Wedge Principle

The lattice exhibits elastic properties up to the rupture point ($E_{\text{rupture}} \approx 147 \text{ GeV}$). By applying a coherent field with energy just below this threshold, it is possible to induce a "phase transition" where the lattice locally expands without collapsing.

3. Operating Parameters

3.1 Resonant Frequency

To oscillate the lattice bonds, a frequency derived from the base mass of 82 GeV is required:

3.2 The Expansion Window

- **Plasticity Threshold:** $> 82.67 \text{ GeV}$. (Below this limit, the lattice is rigid).
- **Critical Point:** 147.44 GeV . (Here, collapse $R=2.0$ occurs).
- **Operating Point: 145.0 GeV .**
 - At this point, the lattice is maximally expanded (in a state of "super-plasticity").
 - The distance between clusters theoretically increases towards infinity (divergence on the graph), creating a macroscopic tunnel.

4. Maintenance Mechanism

To maintain an open gap, a **standing wave** with an energy of 145 GeV must be created ahead of the object. This wave acts as an "icebreaker," pushing Higgs clusters aside in front of the object, while the lattice closes elastically behind it.

5. Conclusion

Enlarging the gap is an energy-intensive but physically plausible process. It requires precise energy regulation to avoid exceeding the 147 GeV limit, which would lead to a destructive release of vacuum energy (explosion).